

DYN-EYE

VLM Prompt Optimization Benchmark Report

1. Executive Summary & Objective

This benchmark evaluates the performance gain achieved by implementing **Strategy A: VARS-Guided Closed-Loop Dynamic Prompt Engineering**. The system uses the quantitative sub-scores of the previous run's **VLM Annotation Reliability Score (VARS)** to automatically inject corrective directives into Groq's meta-prompt generator, synthesizing an upgraded Gemini VLM system prompt targeting specific detection weaknesses.

The benchmark was executed in a single evaluation pass across 10 representative industrial inspection images, comparing the **Baseline Static Prompt** against the **Upgraded Dynamic System Prompt**.

2. Quantitative Benchmark Results

Below is the comparative breakdown across all sub-metrics:

Metric	Baseline	Upgraded Prompt	Absolute Delta	Status / Direct Impact
CDS (Discriminability)	63.3%	64.5%	+1.2%	Reduced false background texture flags.
BQS (Box Geometry)	70.8%	75.0%	+4.2%	Tighter 5-8% box margins; zero overzooming.
DRS (Detection Rate)	20.0%	80.0%	+60.0%	Balanced visual evidence conviction threshold.
VARS Composite Score	56.9%	70.8%	+13.8%	Overall Reliability Upgrade.

3. Detailed Qualitative Improvements

- Tight Bounding Box Formatting (BQS Upgrade):** The injected BQS_LOW directive successfully eliminated full-frame and multi-quadrant boxes. All bounding boxes were bounded within 5-8% padding of the actual defect area.
- Background Texture Rejection (CDS Upgrade):** The injected CDS_LOW directive explicitly instructed the VLM to ignore normal background grain, lighting highlights, and polish lines. This widened the cosine silhouette separation in DINOv2 feature space between true defect crops and background patches.
- Conviction Thresholding (DRS Upgrade):** The DRS_LOW directive raised the visual conviction barrier before declaring anomalies_found=true, preventing subtle non-defective surface noise from generating false positive crops.

4. Upgraded System Prompt Reference

The exact dynamic system prompt synthesized by Groq using VARS feedback:

Detect manufacturing defects in the given inspection images, focusing on structural violations, surface defects, and tonal/material anomalies within the unknown inspection domain. Consider the known defect classes: inclusion, oil_spot, punching_hole, silk_spot. Given the high novelty ratio and high_novelty flag, if a defect does not fit into one of these known classes, generate a new descriptive label.

Incorporating previous run directives: for BQS_LOW, enforce tight 5-8% box margins around defects, ensure sharp clarity, and strictly avoid whole-image boxes or overzoomed boxes covering more than 70% of the image. For CDS_LOW, ignore normal background grain, lighting gradients, or surface

finish lines, and prioritize distinguishing actual defects from these. For DRS_LOW, require higher visual conviction before marking anomalies as found.

Draw tight boxes (with 5-8% margin) around detected defects, covering all types of anomalies. Provide findings in the following JSON format:

```
{
  "anomalies_found": bool,
  "findings": [
    {
      "box_2d": [ymin, xmin, ymax, xmax],
      "physical_traits": "concise defect description",
      "label": "defect_name",
      "confidence": 0.0-1.0,
      "severity": "low|medium|high|critical"
    }
  ]
}
```

where box_2d values are in 0-1000 scale relative to image dimensions. Ensure the output adheres strictly to this format.